Scalability & Future Plan

Date	13 JULY 2026
Team ID	XXXX
Project Name	Comprehensive measure of well-being
Maximum Marks	1 Mark

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

1. Current System Limitations

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution

1	User Load	Limited to low-volume local development connections via	Wrap the Flask application block inside a multi-worker Gunicorn / uWSGI server

This captures the operational constraints of your current lightweight, flat-file architecture.

S. No	Limitation	Impact	Priority to Address
1	Single-Process Flask Runtime	The built-in Werkzeug development server blocks when multiple evaluators send concurrent requests, risking timeout errors.	High
2	In-Memory Volatile Processing	No persistence layer is integrated; past predictions and scenario simulation metrics vanish the moment the browser tab is closed.	Medium
3	Static Model Artifact (HDI.pkl)	The regression model is hard-coded. Updating it with new country rows requires manually rebuilding the file and restarting the backend server process.	Medium

2. Scalability Plan

This maps out how to evolve the platform away from its local workspace architecture into a production-grade infrastructure solution.

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
		an Ngrok edge node.	container managed behind an Nginx reverse proxy layer.
2	Data Storage	No server footprint; reads directly from localized HDI.csv files on startup.	Integrate an embedded SQLite database engine for light historical query tracking before moving workloads to a managed PostgreSQL cluster.
3	Performance	Blocked by synchronous NumPy/Pandas calculations handled on a single system thread.	Move incoming payload tasks into asynchronous processing queues using Celery workers backed by an in-memory Redis cache block.
4	Security	Relying entirely on simple local Flask environment route validation catches.	Restructure endpoints under an API Gateway module enforcing JWT user authentication strings and cross-origin (CORS) security headers.

3. Future Roadmap

This structures your multi-tiered development timeline for subsequent release cycles beyond the current core configuration.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Low-Footprint Query Persistence	1 Month Out	Enables long-term session tracking, allowing policymakers to log and audit historical simulation data over time.
Phase 3	Multi-Language Localization Grid	3 Months Out	Expands accessibility, permitting cross-border global development teams to engage with the dashboard natively.
Phase 4	Time-Series Trajectory Models	6 Months Out	Elevates analytical precision by shifting from single point predictions to continuous multiyear development forecasts.